Solution:

1. **D. 24**
Note:
$$\text{og}_a b = \frac{\log_b b}{\log_b a} = \frac{1}{\log_b a}$$
Thus,
$$\log_a 2 + \log_a 3 + \log_a 4 = 1$$
$$\log_a 24 = 1$$
$$a = 24$$

2. **C. 0.25**
$$\log\left[k(x+y)\right] = \frac{1}{2}(\log x + \log y)$$
$$2\log\left[k(x+y)\right] = \log x + \log y$$
$$\log\left[k(x+y)\right]^2 = \log(xy)$$
$$\left[k(x+y)\right]^2 = xy$$
$$k^2(x+y)^2 = xy$$
$$x^2 + 2xy + y^2 = \frac{xy}{k^2}$$
$$x^2 + y^2 = \frac{xy}{k^2} - 2xy$$
$$x^2 + y^2 = \left(\frac{1}{k^2} - 2\right)xy$$
So, from $x^2 + y^2 = 14xy$:
$$14xy = \left(\frac{1}{k^2} - 2\right)xy$$
$$14 = \frac{1}{k^2} - 2$$
$$\frac{1}{k^2} = 16$$
$$k = \pm\frac{1}{4}$$

3. **C. 2**
If the roots are 3 and 2, then:
$$Ax^2 + Bx + C = 0$$
$$(x-3)(x-2) = 0$$
$$x^2 - 5x + 6 = 0$$
$$A + B + C = 1 - 5 + 6$$
$$A + B + C = 2$$

4. **D. 2/3**
$$2\log_a x - \log_a(x-1) - \log_a(x-2) = 0$$
$$\log_a \frac{x^2}{(x-1)(x-2)} = 0$$
$$\frac{x^2}{x^2 - 3x + 2} = a^0$$
$$x = \frac{2}{3}$$

5. **B. 90**
$15 = 5\cdot 3$
$18 = 6\cdot 3$
LCM $= 5\cdot 6\cdot 3 = 90$

6. **B. 25 + k**
$p - q = 5$
By squaring both sides:
$$(p-q)^2 = 5^2$$
$$p^2 - 2pq + q^2 = 25$$
$$p^2 + q^2 = 25 + 2pq$$
$$p^2 + q^2 = 25 + 2\left(\frac{k}{2}\right)$$
$$p^2 + q^2 = 25 + k$$

7. **D. 4**
$$x^2 + 2(k+2)x + 9k = 0$$
For equal roots $B^2 - 4AC = 0$
$$x^2 + 2(k+2)x + 9k = 0$$
$$B^2 - 4AC = 0$$
$$(2k+4)^2 - 4(1)(9k) = 0$$
$$4k^2 - 20k + 16 = 0$$
$$k_1 = 4; k_2 = 1$$

8. **A. 3**
$$3x^2 - kx + x - 7k = 0$$
At x = 3,
$$3(3)^2 - k(3) + 3 - 7k = 0$$
$$k = 3$$

9. **B. 1:2**
$$\frac{1}{x} : \frac{1}{y} \rightarrow \frac{1}{x} \div \frac{1}{y} = \frac{y}{x}$$
$$2x + 4y = 7x - 6y$$
$$y(4+6) = x(7-2)$$
$$\frac{y}{x} = \frac{1}{2}$$
$$\frac{1}{x} : \frac{1}{y} \rightarrow 1:2$$

10. **B. xy**
$$\frac{(xy)^n (a-b)^{-1} (a-b)^{2n-1}}{\left[xy(a-b)^2\right]^{n-1}}$$
$$\frac{(xy)^n \left(\frac{1}{a-b}\right)(a-b)^{2n}(a-b)^{-1}}{(xy)^n (xy)^{-1} (a-b)^{2n-2}}$$
$$\frac{\frac{(a-b)^{2n}}{(a-b)^2}}{\left(\frac{1}{xy}\right)(a-b)^{2n}(a-b)^{-2}}$$
$$\frac{xy(a-b)^2}{(a-b)^2} = xy$$

11. **B. 12 days**
Wife's day off: 4th day
Husband's day off: 6th day
Getting the least common multiple of 4 and 6
LCM = 12

12. **A. 43,750x^8**
Binomial Expansion:
Total number of terms = n + 1
$(x^2 - 5)^8$; n = 8
Number of terms = 9
Middle term = 5th term
Nth Term = ${}_nC_{r-1} a^{n-r+1} b^{r-1}$
5th Term = ${}_8C_{5-1}\left(x^2\right)^{8-5+1}(-5)^{5-1}$
5th Term = $43,750x^8$

13. **A. 2**
Applying Factor Theorem
If $\frac{f(x)}{x-k}$; $\text{Remainder} = \underset{x\to k}{f(x)} = 0$
Then,
$$\frac{kx^2 - 6x + 2kx - 12}{x-3}$$
$$\text{Remainder} = f(x) = 0$$
At x = 3
$$kx^2 - 6x + 2kx - 12 = 0$$
$$k(3)^2 - 6(3) + 2k(3) - 12 = 0$$
$$k = 2$$

14. **C. – 5**

Applying Remainder Theorem

If $\frac{f(x)}{x-k}$; $\underset{x\to k}{\text{Remainder}} = f(x)$

Then,

$$\frac{x^4+ax^3+5x^2+bx+6}{x-2}$$

Remainder = f(x) = 16 at x = 2

$$x^4+ax^3+5x^2+bx+6=16$$
$$(2)^4+a(2)^3+5(2)^2+b(2)+6=16$$
$$8a+2b=-26 \rightarrow \text{eqn } 1$$

When,

$$\frac{x^4+ax^3+5x^2+bx+6}{x+1}$$

Remainder = f(x) = 10 at x = -1

$$x^4+ax^3+5x^2+bx+6=10$$
$$\left(\begin{matrix}(-1)^4+a(-1)^3+5(-1)^2\\+b(-1)+6\end{matrix}\right)=10$$
$$-a-b=-2 \rightarrow \text{eqn } 2$$

Solving for a and b:

a = -5 and b = 7

15. **C. – 280(a^4)(x^3)**

$$(a-2x)^7$$

rth Term = ${}_nC_{r-1}a^{n-r+1}b^{r-1}$

4^{th} Term = ${}_7C_{4-1}a^{7-4+1}(-2x)^{4-1}$

4^{th} Term = $-280a^4x^3$

16. **B. 25,344x^9**

rth Term = ${}_nC_{r-1}a^{n-r+1}b^{r-1}$

rth Term = ${}_{12}C_{r-1}\left(x^2\right)^{12-r+1}\left(\frac{2}{x}\right)^{r-1}$

Considering the terms w/ x for x^9:

$$\left(x^2\right)^{12-r+1}\left(x^{-1}\right)^{r-1}=x^9$$
$$2(12-r+1)+(-1)(r-1)=9$$
$$24-2r+2-r+1=9$$
$$r=6$$

6^{th} Term = ${}_{12}C_{6-1}\left(x^2\right)^{12-6+1}\left(\frac{2}{x}\right)^{6-1}$

6^{th} Term = $25,344x^9$

17. **A. 1**

$$x=(\log_c a)(\log_a b)(\log_b c)$$
$$x=\left(\frac{\log a}{\log c}\right)\left(\frac{\log b}{\log a}\right)\left(\frac{\log c}{\log b}\right)=1$$

18. **B. 256**

$$(x+2y-z)^8$$

Sum of Coefficients = (let all variables to 1)^n

Sum of Coefficients = $(1+2-1)^8$

Sum of Coefficients = 256

19. **B. e and e^5**

$$(\ln x)^2-2\ln x^3=-5$$
$$(\ln x)^2-6\ln x=-5$$

Let A = ln x

$$A^2-6A+5=0$$
$$A_1=5 \;;\; A_2=1$$

At A = 5,	At A = 1
$\ln x=5$	$\ln x=5$
$x=e^5$	$x=e$

20. **A. 21**

$$x:y:z=2:5:7$$
$$\frac{x}{y}=\frac{2}{5}$$
$$5x-2y=0 \rightarrow \text{eqn } 1$$
$$\frac{y}{z}=\frac{5}{7}$$
$$7y-5z=0 \rightarrow \text{eqn } 2$$
$$4x-y+2z=51 \rightarrow \text{eqn } 3$$

x = 6, y = 15 and z = 21

21. **C. – 66339/128a^11**

Binomial Expansion $\left(\frac{1}{2}-3\right)^{16}$

rth Term = ${}_nC_{r-1}a^{n-r+1}b^{r-1}$

where, n = 16

6th Term = ${}_{16}C_{6-1}\left(\frac{1}{2a}\right)^{16-6+1}(-3)^{6-1}$

6th Term = $-\frac{66339}{128a^{11}}$

22. **D. y = xb^{2x}**

$$\log_b y=2x+\log_b x$$
$$\log_b y-\log_b x=2x$$
$$\frac{\log_{10} y}{\log_{10} b}-\frac{\log_{10} x}{\log_{10} b}=2x$$
$$\log_{10} y-\log_{10} x=2x\log_{10} b$$
$$\log_{10}\frac{y}{x}=\log_{10} b^{2x}$$
$$y=xb^{2x}$$

23. **A. 20**

$$(x+y+z)^5$$

Trinomial Expansion

rth Term = $\frac{n!}{p!q!s!}(a)^p(b)^q(c)^s$

where, n = p + q + s

Identifying the term with xy^3z:

$$\frac{5!}{p!q!s!}(x)^p(y)^q(z)^s$$
$$x^p=x; p=1$$
$$y^q=y^3; q=3$$
$$z^s=z; s=1$$

Thus for coefficient, $\frac{5!}{1!3!1!}=20$

24. **A. 78**

Binomial Expansion $\left(x^a+y^b\right)^n$

Sum of exponents: $n(n+1)\left(\frac{a+b}{2}\right)$

Then, $\left(x^2+\frac{2}{x}\right)^{12}$

$$12(12+1)\left(\frac{2+-1}{2}\right)=78$$

25. **A. n(n + 1)^2**

$$\frac{[(n+1)!]^2}{n!(n-1)!}=\frac{[(n+1-1)!(n+1)]^2}{n!(n-1)!}$$
$$=\frac{(n!)^2(n+1)^2}{n!(n-1)!}$$
$$=\frac{(n+1)^2(n)(n-1)!}{(n-1)!}$$
$$=n(n+1)^2$$

"Successful people do what unsuccessful people are not willing to do."